Maddipatla Venkata Sai Gowtham

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.64
Sri Chaitanya Junior Kalasala	Class XII (Telangana State Board of Intermediate Education)	2021	98.80%
Sri Aurobindo Centenary School	Class X (Board of Secondary Education Andhra Pradesh)	2019	10.00%

SKILLS

- **Hardware Description Languages:** Verilog
- **Programming Languages :** C, Python
- **VLSI Tools :** Xilinx Vivado , Sentaurus TCAD , Cadence Virtuoso , LTspice

RELEVANT COURSES

- **Digital Domain :** Digital IC Design, CMOS Digital Design, Digital IC Design Lab
- **Analog Domain** : Analog IC Design, Analog IC Design Lab
- **Devices Domain :** Semiconductor Devices and Modelling, Memory Device Technology for AI/ML, Nanoelectronics: Principles and Devices

M.TECH THESIS

Title: Device Design and Simulations of InP/InGaAs HEMT for low noise and T-Hz frequency applications.

Supervisor: **Dr. Kaushik Nayak**, Associate Professor, Electrical Engineering, IIT Hyderabad

Thesis Description: InP/InGaAS HEMT device design and simulation using Sentaurus TCAD tool to evaluate DC and high frequency performance for low noise sub-mm/T-Hz frequency operation at low cryogenic temperature(10K-50K)

PROJECTS

Op-Amp Based Voltage-Mode Bandgap Reference

- Designed a voltage-mode BGR using an op-amp feedback loop to generate a stable 1.2 V reference across PVT variations.
- Implemented a start-up circuit to eliminate the zero-current state and ensure reliable circuit initialization across operating corners.

CMOS Two-Stage OTA Design

- Designed and simulated a two-stage CMOS OTA using GPDK90nm technology in Cadence Virtuoso.
- Analyzed its DC, AC, and PVT performance for reliable analog operation.

Implementation of Dual-Clock Asynchronous FIFO using RTL

- Designed and implemented an asynchronous FIFO in RTL with independent read and write clock domains.
- Utilized Gray-code encoding for pointer synchronization to minimize multi-bit transitions and reduce metastability risks.

EXTRACURRICULAR ACTIVITIES

- Nature Club Member, NIT Andhra Pradesh - Organized and conducted an Ideathon event.
- ECE Association Member, NIT Andhra Pradesh.

CERTIFICATIONS

- NPTEL: Data Science for Engineers

HOBBIES

Hobbies

- Playing Cricket
- Watching Movies